

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set: A

Examination: Semester Final
Duration: 1 Hour 45 Minutes

Semester: Fall 2022
Full Marks: 45

CSE 220: Data Structures

Answer all of the following questions.
Figures in the right margin indicate marks.

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1. Suppose you are given a string with duplicate characters. Your task is to complete the following recursive function `remove_duplicate()` that takes a string and an index and removes all the adjacent duplicate characters and returns the new string. 10

CO4

[Write the function only for adjacent duplicates]

Sample Input	Sample Output
<code>remove_duplicate('CSSSE', 0)</code>	CSE
<code>remove_duplicate('Programming is fun', 0)</code>	Programing is fun

(Python signature)

```
def remove_duplicate(st, idx):  
    # write your code here
```

(Java signature)

```
public String remove_duplicate(String st, int idx) {  
    // write your code here  
}
```

2. A. The capacity of a circular queue is 6. Getting the alphabet means `enqueue(alphabet)`. '`&`' means `dequeue()`, '`$`' means `peek()`. You have to write the output of the circular queue after each operation. The front index of the queue is 4. 5

CO3

h t z & \$ f & & e \$ k & \$ m & t

Also find out which element is the front and which element is the rear/back of Queue after Last operation

CO1

B. Complete the following functions for linked list based queue

Python	Java
def dequeue(head): # To do	public void dequeue(Node head) { # To do }
def enqueue(head, val): # To do	Public void enqueue(Node head, int val) { # To do }

2

3

3.

CO2

Given a linked list containing English words. You have to count how many times each word is occurring.

a) If you perform a linear search, what is the time the algorithm will take? 1

b) In order to speed up the process key index searching is proposed. What is the limitation of using key index search for the problem mentioned above? 1

c) To address the problem a modified version is proposed. For each English word, an integer will be generated through some function. The integer will be such that it is a valid index of an auxiliary array. If A = 1, B = 2, ... Z = 26, assuming the words are all in caps, the function 6

f(word) = [(multiplication of the numbers associated with each letter)] % 10.

For example, if CAT is the word, $f(CAT) = (3*1*20)\% 10 \Rightarrow 0$. Inside index number 0 of the auxiliary array, it will be 1. Based on the information and your concept of an auxiliary array key index searching, **complete** the method. The createAUX method takes the head of a singly linked list. Return the array at the end. **The length of the array is hidden in the question.**

def createAUX(head):

 return aux #aux is the auxiliary array

d) What possible problem could arise in part c? 2

4.
CO2

A. Consider the following numbers:

21, 9, 15, 30, 35, 25, 23, 17, 4, 11, 27

- a) Draw Binary Search Tree by inserting the above numbers as keys of the nodes from left to right. 2
- b) Show the pre-order, post-order, and in-order traversal sequence of the tree. 3
- c) Perform the following operations step by step on the Binary Search Tree you created in part 'a' 3
- i. Delete node with the key of value 30 with the help of its predecessor.
 - ii. Delete node with the key of value 21 with the help of its successor.

B.

Consider the following Adjacency Matrix representation of an undirected graph

	A	B	C	D	E	F
A	0	0	0	0	0	1
B	1	0	1	1	0	1
C	0	1	1	1	0	1
D	0	1	1	0	1	1
E	0	0	1	1	0	1
F	1	0	1	1	0	0

- a) Is the above Adjacency Matrix a correct representation for an undirected graph? If not, fix the matrix without deleting and decreasing weight of any of the edges. 3
- b) Construct an undirected graph from the correct Adjacency Matrix. 2
- c) Represent the above graph with the help of an Adjacency List. 3